



DATA & AI IN ECONOMICS

Loan Default Risk

Causal Inference, Borrower Segmentation, and Supervised Prediction

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1 Motivation & Research Question

Anchoring loan default prediction in an economic mechanism, not just a classification task



The Economic Mechanism

Higher debt-to-income (DTI) ratio → larger repayment burden relative to income.

Tighter liquidity → income shocks (job loss, rate hikes) are harder to absorb.

Policy relevance: responsible-lending rules, over-indebtedness regulation, and transparent automated credit decisions.

Hypotheses

- H1: High DTI raises default probability
- H2: The causal effect varies by borrower segment
- H3: Clustering improves economic interpretability

RESEARCH QUESTION

Does a higher debt-to-income ratio causally increase the probability of loan default?



Beyond Prediction

Goes past 'can we predict?' to isolate a causal driver of risk



Stakeholders

Banks, regulators, credit-risk teams, and automated lending systems

2 Dataset Overview & Data Quality Plan

Full reproducibility reference, plus diagnostics paired with concrete mitigation strategies



Source & License

Dataset	Loan Default Prediction Dataset
Author	NIKHIL (Kaggle), 2023
License	CC0 1.0 — Public Domain
Rows / Vars	255,347 borrowers / 18 fields
Target	Default (binary)
Treatment	DTIRatio → HighDTI

kaggle.com/datasets/nikhil1e9/loan-default

Data Quality → Mitigation



Missing values



Median (numeric) / most-frequent (categorical) imputation



Class imbalance



Stratified split + class_weight='balanced' + SMOTE



Outliers



IQR-based clipping / winsorization



Scale differences



Standardization before LR, PCA, K-Means



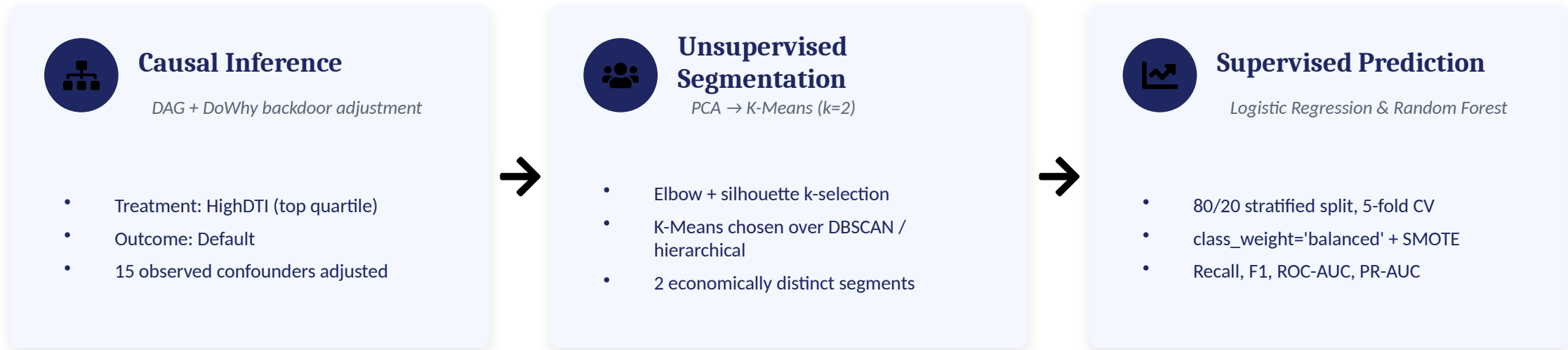
Data leakage



Drop LoanID identifier before modeling

3 Methodology: An Integrated Workflow

Causal inference, clustering, and prediction are connected, not run in parallel



Methodological Synthesis — How the Blocks Connect

- Default rates & profiles compared across clusters (economic interpretation)
- HighDTI effect re-estimated within each cluster (causal heterogeneity)
- Cluster membership tested as a feature in the prediction model

4 Main Results

Causal effect, borrower segments, and best-performing predictive model

CAUSAL EFFECT (ATE)

+0.88 pp

High DTI borrowers default ~0.88 percentage points more often, after adjusting for income, credit score, employment and 12 other confounders.



Default Rate by Segment

Cluster 0 (constrained, n=87,979)



Cluster 1 (stable, n=167,368)



Best Model: Logistic Regression (Balanced)

0.762

ROC-AUC

69.8%

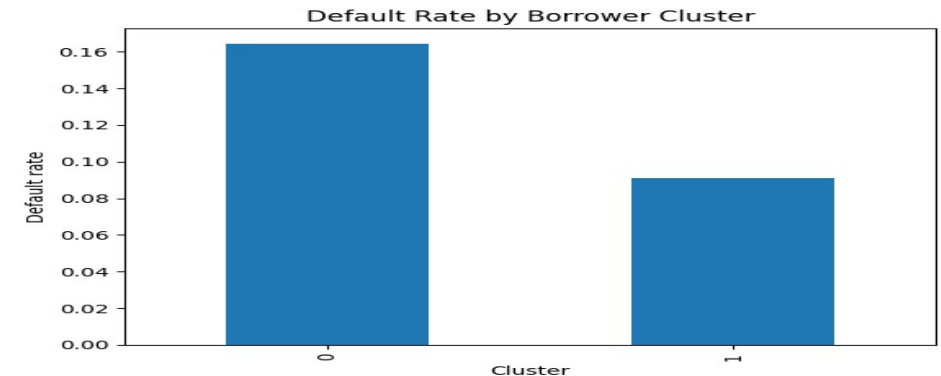
Recall

Outperforms Random Forest Balanced (AUC 0.754) and SMOTE variant on recall & AUC

Heterogeneous Causal Effect & Borrower Profile

Cluster	Default Rate	ATE of HighDTI	Profile
0 — Constrained	16.43%	0.0092	Lower income, high loan-to-income (4.29x)
1 — Stable	9.08%	0.0085	Higher income, low loan-to-income (1.07x)

Effect is positive in both segments — financial leverage raises default risk consistently, slightly more so for constrained borrowers.



5 Limitations & Synthesis

What the integrated framework delivers, and where caution is warranted



Limitations

Causal assumptions:

Unconfoundedness, overlap & SUTVA may be violated by unobserved factors (financial literacy, local shocks)

Dataset realism:

Kaggle data may not fully represent a real bank portfolio — treat as methodological demonstration

Measurement:

Education, employment type are coarse proxies for borrower stability

External validity:

Findings may not generalize across countries, lenders, or time

Predictive trade-off:

High-recall models generate many false positives



Conclusion



High debt burden causally raises default risk (+0.88 pp ATE), supporting H1.



Two interpretable borrower segments emerge — constrained vs. stable — with materially different risk and leverage profiles (H3 supported).



The causal effect is positive in both segments, confirming the mechanism is robust across borrower types (H2 supported).



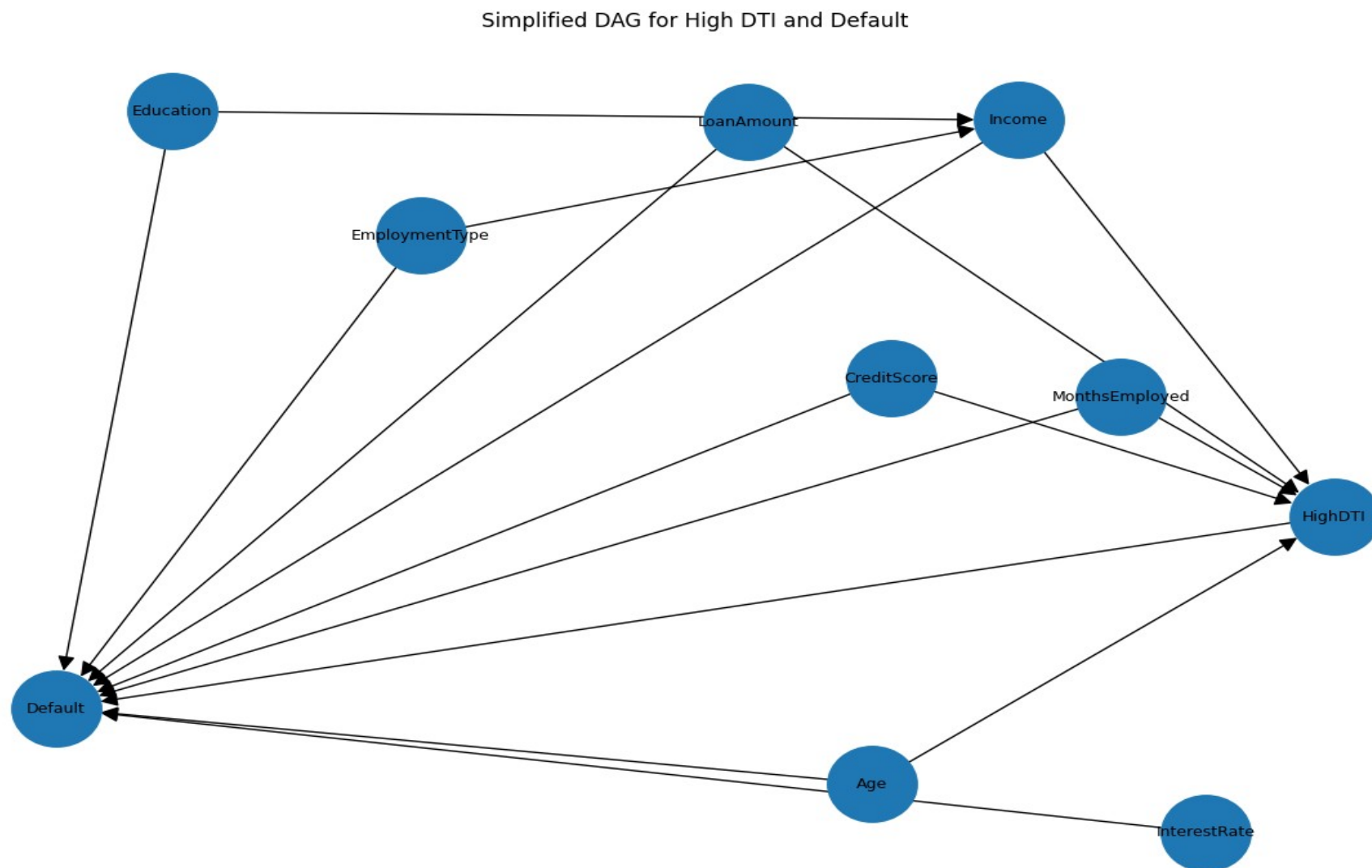
Cluster membership adds no numerical predictive lift, but is the key driver of economic interpretability — the methods complement, not duplicate, each other.

References

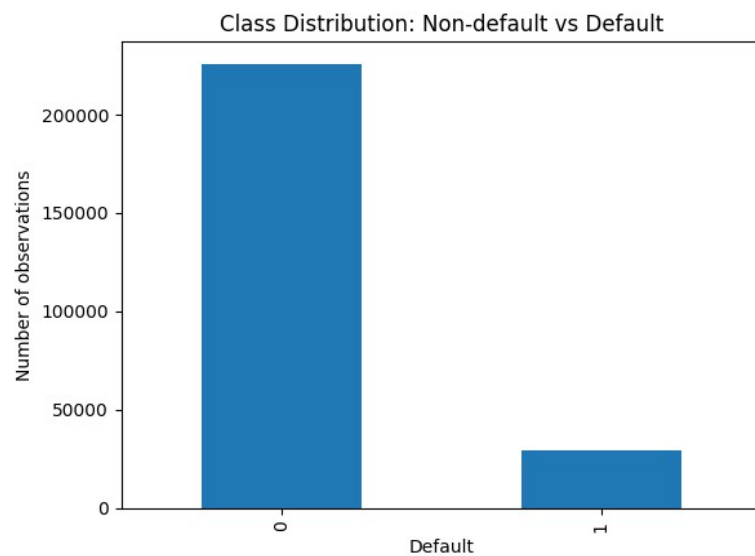


- 1 NIKHIL (2023). Loan Default Prediction Dataset. Kaggle. <https://www.kaggle.com/datasets/nikhil1e9/loan-default/data>. License: CC0 1.0 Universal Public Domain Dedication. Accessed 29 June 2026.
- 2 Sharma, A., & Kiciman, E. (2020). DoWhy: An End-to-End Library for Causal Inference. arXiv preprint.
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- 4 Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825-2830.
- 5 Chawla, N. V. et al. (2002). SMOTE: Synthetic Minority Over-sampling Technique. Journal of Artificial Intelligence Research, 16, 321-357.

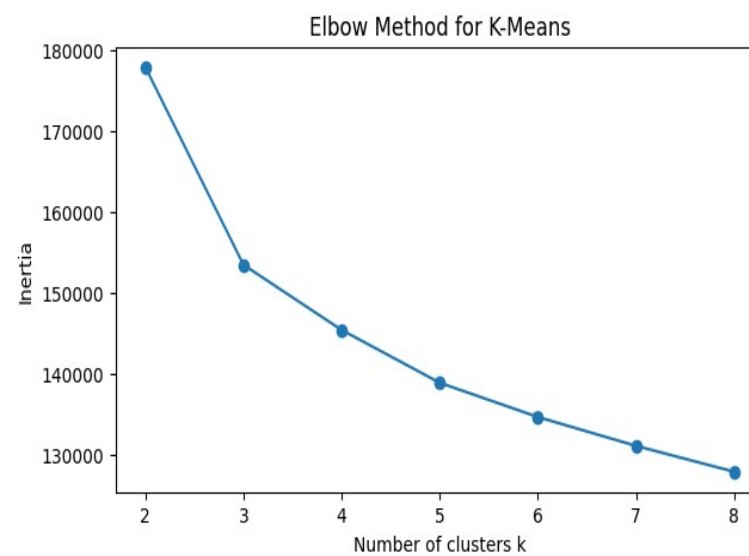
Causal DAG: Backdoor Adjustment Structure



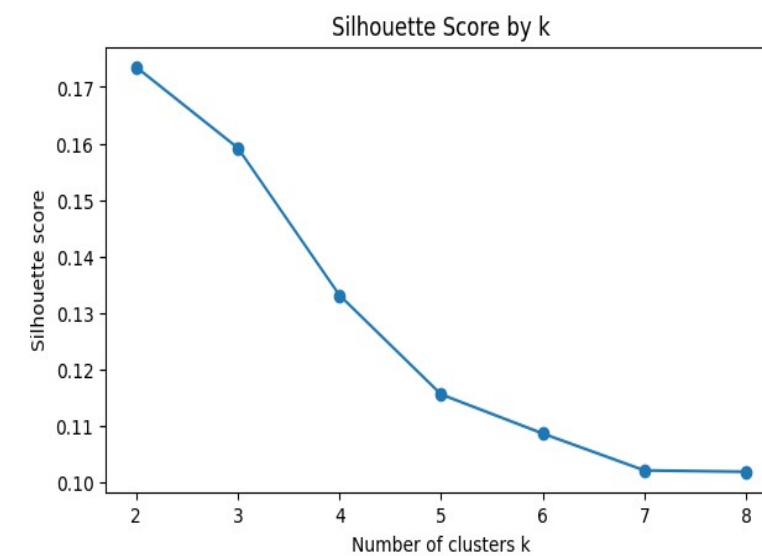
Class Imbalance & K-Selection Diagnostics



Severe class imbalance ($\approx 11\%$ defaults) motivates stratified splits, class weighting, and SMOTE.



Elbow inertia and silhouette score both support $k=2$ as a stable, interpretable choice.



Full Supervised Model Comparison

Model	Recall	Precision	F1	ROC-AUC	Avg. Precision
Logistic Regression Balanced	0.6980	0.2283	0.3441	0.7615	0.3362
Logistic Regression + SMOTE	0.6936	0.2289	0.3442	0.7614	0.3360
Random Forest Balanced	0.4814	0.2958	0.3664	0.7538	0.3229

Logistic Regression Balanced achieves the highest Recall and ROC-AUC; Random Forest achieves the highest Precision and F1. Given the cost asymmetry of missed defaults, Recall and ROC-AUC are prioritized.

Effect of Adding Cluster Membership as a Feature

Specification	Recall	Precision	F1	ROC-AUC
Without cluster	0.6980	0.2283	0.3441	0.7615
With cluster	0.6980	0.2283	0.3441	0.7615

Borrower Cluster Economic Profiles

Cluster	Size	Default Rate	Avg. Income	Avg. Loan Amount	Avg. DTI	Avg. Loan/Income
0 — Constrained	87,979	16.43%	\$43,859.67	\$162,291.69	0.5001	4.2861
1 — Stable	167,368	9.08%	\$102,810.69	\$109,331.65	0.5003	1.0691

